Matt Kuehr

Email : mck0063@auburn.edu
https://www.linkedin.com/in/matthew-kuehr/
Mobile : 1-205-478-9916
https://github.com/MattKuehr

Education

- **Auburn University** — Auburn, Alabama
  M.S. Data Science — *Expected May 2027*
- **Auburn University** — Auburn, Alabama
  B.S. Applied Mathematics; GPA: 3.97 — *May 2026*
- **Auburn University** — Auburn, Alabama
  Undergraduate Certificate - Artificial Intelligence Engineering; GPA: 4.0 — *May 2026*

Experience

- **Hadrian Automation** — Torrance, California
  Engineering Intern — *May 2026 - Present*
  - **OCR Dataset Curation**: Wrote a Python software tool to facilitate ingestion and stratification of large OCR datasets. Explored optimality in stratification decisions by analyzing embedding space toplogy using nonlinear dimensionality reduction techniques such as t-SNE and UMAP in conjunction with more traditional PCA-based methods.
- **Auburn University Department of Mathematics and Statistics** — Auburn, Alabama
  Undergraduate Research Assistant — *January 2024 - Present*
  - **Time Series Classification With Transformers**: Applied transformer-based models to time series classification problems in the aerospace domain. Adjusted the training regimen and model architecture to address issues raised by data sparsity from both deterministic and stochastic origins.
  - **Neural Network Surrogate Modeling For Flight Behavior**: Trained a neural network in PyTorch on aircraft geometries to predict the aircraft's lift, drag and moment coefficients. Applied random search in hyperparameter space to optimize the architecture of the neural network surrogate model. Results were presented at the AIAA SciTech conference in January 2026.
  - **From-Scratch Neural Network Implementation**: Implemented a neural network and automatic differentiation engine from scratch using only the NumPy library. On our in-house missile classification dataset, the network performed competitively and sometimes exceeded the metrics of Scikit-Learn and PyTorch implementations.
- **Auburn University Department of Mathematics and Statistics** — Auburn, Alabama
  Undergraduate Research Fellow — *August 2025 - December 2025*
  - **Numerical Methods for Band Structure Calculation**: Developed an algorithm in Python to solve for the band structure of 1D non-Hermitian topological photonic crystals. NumPy was utilized to numerically calculate eigenvalues and SciPy was used for root finding. The algorithm also applied the high-performance computing library Numba to leverage CPU parallelization and just-in-time (JIT) compilation.
  - **Dataset Curation**: Generated a dataset of over 2000 photonic crystal band structures using the previously described band structure calculation algorithm. By varying several geometric parameters in the generation process, a robust and diverse dataset was created for use in downstream machine learning applications.
  - **Automatic Band Detection**: Developed an algorithm in Python to automatically detect bands in images of photonic crystal band structures. The algorithm applied K-D trees and heuristic approaches to the corpus of points recovered by the band structure calculation algorithm to identify independent bands within the images.
- **Hadrian Automation** — Torrance, California
  Engineering Intern — *May 2025 - August 2025*
  - **Title Block Detection**: Developed a computer vision algorithm to detect the position of bounding boxes on engineering CAD drawings. The detection algorithm was a custom pipeline built on top of standard computer vision methods in the OpenCV Python library.
  - **Vision Language Model Finetuning**: Finetuned a Vision Language Model (VLM) to extract critical information from title blocks on engineering CAD drawings. The model was finetuned and deployed on the AWS Sagemaker cloud computing framework. Used PyTorch and the Hugging Face transformers framework to implement a LoRA-based SFT pipeline.

Papers

- **Surrogate Modeling of Flight Behavior With Deep Neural Networks (SciTech 2026)**: Primary author of a paper exploring the efficacy of using deep neural networks to predict aerodynamic forces and moments from aircraft orientation and geometry.

Software Skills

- **Languages**: Bash, C++, Python, SQL, TypeScript
- **Frameworks**: Hugging Face, NumPy, PyTorch, Scikit-Learn, SciPy
- **Developer Tools & Systems**: AWS, Docker, Git, Linux, PostgreSQL, Shell Scripting

Awards and Achievements

- **Andrew C. Connor Memorial Award**: Awarded by the Department of Mathematics and Statistics for creativity and academic excellence during the 2024-2025 academic year.
- **Business Analytics Case Competition (First Place)**: Won the 2025 Business Analytics Case Competition organized by the Auburn University Harbert College of Business and sponsored by The Home Depot and Regions Bank.
- **COSAM Dean's Medal**: Departmental recipient of the College of Sciences and Mathematics Dean's Medal in 2026 for the Department of Mathematics and Statistics.
- **Jack and Jane Brown Endowed Scholarship Recipient**: Awarded for academic excellence during the 2024-2025 academic year.

References

- **Dr. Mark Carpenter — Auburn University**: carpedm@auburn.edu
- **Dr. Roy Hartfield — Auburn University**: hartfrj@auburn.edu